ENGINEERING ECONOMY

LECTURE 10

PROF.DR. YASEMİN ASU ÇIRPICI, MU, DEPT. OF ECONOMICS

Rate of Return Evaluation Using AW

- Comparing alternatives by ROR analysis method always leads to the same selection as PW, AW, FW analysis method, whether ROR is determined using PW-based, AW-based or FW-based.
- All methods must consider equal service principle
- For AW-based ROR method there are two ways to perform the evaluation for *i* for the incremental cash flow series:

  1 The incremental cash flows must be evaluated over the LCM of lives (results in *i*)

  2 Set the difference between AW relations for each alternative to zero

  $0 = AW_B - AW_A$ (resulting $i = i^*$)

  This way requires less computational effort

EXAMPLE

- Our company has the option of supplying an equipment from two different suppliers - a U.S. firm (A) and an Asian firm (B). Estimates of this equipment from vendors A and B are given below. Decide which vendor should be selected using AW-based incremental ROR method and MARR of 15%. (external investment rate is 12.65%)

	A	B
Initial cost, $	-8,000	-13,000
Annual costs, $	-3,500	-1,600
Salvage Value, $	0	2,000
Life, years	10	5

Year	Cash Flow A	Cash Flow B	Incremental Cash Flow (B-A)
0	-8,000	-13,000	-5,000
1-5	-3,500	-1,600	1,900
5		2,000 -13,000	-11,000
6-10	-3,500	-1,600	1,900
10		2,000	2,000

- Using the incremental cash flows: it must be evaluated over 10 years
- Set the difference between AW relations for each alternative to zero using 5 and 10 years
- $0 = AW_B - AW_A$

1. $\Delta i^* = 12.65\%$ is found from the following equation using LCM years and incremental cash flows:

$$\begin{aligned} 0 &= -5,000(A/P, \Delta i, 10) + 1,900 + 2,000(A/F, \Delta i, 10) \\ &- 11,000(P/F, \Delta i, 5)(A/P, \Delta i, 10) \end{aligned}$$

2. Same $\Delta i^* = 12.65\%$ can be also found from:

$$\begin{aligned} 0 &= AW_B - AW_A \\ &= -13,000(A/P, \Delta i, 5) - 1,600 + 2,000(A/F, \Delta i, 5) \\ &- [-8,000(A/P, \Delta i, 10) - 3,500] \end{aligned}$$

Since $\Delta i^* = 12.65\% < 15\%$, the additional \$5,000 of investment is not justified, and alternative A is chosen.

- Note that external interest rate is calculated from solving:

Incremental Cash Flows (B − A), LCM = 10 years

- $t = 0$: $-5{,}000$
- $t = 1, 2, 3, 4$: $+1{,}900$
- $t = 5$: $+1{,}900 - 11{,}000 = -9{,}100$
- $t = 6, 7, 8, 9$: $+1{,}900$
- $t = 10$: $+1{,}900 + 2{,}000 = +3{,}900$

NPV = 0

$$
\begin{aligned}
0 = &-5{,}000 \\
&+ \frac{1{,}900}{(1+i)^1} + \frac{1{,}900}{(1+i)^2} + \frac{1{,}900}{(1+i)^3} + \frac{1{,}900}{(1+i)^4} \\
&- \frac{9{,}100}{(1+i)^5} \\
&+ \frac{1{,}900}{(1+i)^6} + \frac{1{,}900}{(1+i)^7} + \frac{1{,}900}{(1+i)^8} + \frac{1{,}900}{(1+i)^9} \\
&+ \frac{3{,}900}{(1+i)^{10}}
\end{aligned}
$$

- $i = 12\%$ NPV +137.7
- $i = 13\%$ NPV −71.3

- There is a root between these values.
- If we use numerical methods or Excell we can find the exact value.
- $\Delta i^* \approx 12.65\ \%$

Finding the Incremental IRR (Δi*) Using Goal Seek

- **Set up incremental cash flows (B − A)**
- Use the common study period (LCM = 10 years).
- Enter an initial guess for the interest rate i (e.g., 10%) in a single cell.
- Compute NPV as a function of I
- Use Goal Seek

Year	Incremental Cash Flow
0	−5,000
1–4	+1,900 each year
5	−9,100
6–9	+1,900 each year
10	+3,900

$$\text{NPV}(i) = -5{,}000 + \sum_{t=1}^{10} \frac{CF_t}{(1+i)^t}$$

(Equivalently in Excel: `=NPV(i, CF_1:CF_10) + CF_0`)

Goal Seek Setup

1. **Compute NPV**
   - Calculate NPV as a function of the interest rate i.
2. **Open Goal Seek**
   - Go to **Data → What-If Analysis → Goal Seek**.
3. **Fill in the dialog box**
   - **Set cell:** the cell containing the NPV formula
   - **To value:** `0`
   - **By changing cell:** the cell containing the interest rate i
4. **Run Goal Seek**
   - Click **OK**.
   - Excel iteratively adjusts i until **NPV = 0**.

Exact Excel Formula Syntax (Goal Seek – NPV = 0)

Spreadsheet Layout

- **A2:A12** → Year (0 to 10)
- **B2:B12** → Incremental Cash Flows (B − A)
- **D2** → Interest rate i (initial guess, e.g. `0.10`)
- **D5** → NPV

NPV Formula (Cell D5)

```excel
=NPV($D$2,B3:B12)+B2
```

- `$D$2` → interest rate i
- `B3:B12` → cash flows at years 1–10
- `+B2` → cash flow at year 0 (Excel NPV excludes year 0)

Goal Seek Settings

- Set cell: `D5`
- To value: `0`
- By changing cell: `D2`

After running Goal Seek:

```
D2 = 0.1265   →   Δi* = 12.65%
```

Optional (Alternative Full-PV Formula)

If you prefer a fully explicit formula (no Excel NPV function):

```
=B2/(1+$D$2)^A2 + B3/(1+$D$2)^A3 + B4/(1+$D$2)^A4 + ... + B12/(1+$D$2)^A12
```

EXAMPLE

- Two mutually exclusive alternatives are being considered.

Item	Alternative A	Alternative B
Initial cost ($)	−9,000	−14,000
Annual operating cost ($/year)	−3,200	−1,900
Salvage value ($)	0	3,000
Useful life (years)	12	6

The **MARR = 14% per year**.

Step 1: Least Common Multiple (LCM) of Lives

- Life of A = 12 years
- Life of B = 6 years
- **LCM = 12 years**

Thus:

- Alternative A is used once,
- Alternative B is repeated twice.

Step 2: Construct Incremental Cash Flows (B − A)

Cash Flow Table

Year	A ($)	B ($)	Incremental (B − A)
0	−9,000	−14,000	−5,000
1–6	−3,200	−1,900	+1,300
6	0	+3,000 −14,000	−11,000
7–12	−3,200	−1,900	+1,300
12	0	+3,000	+3,000

Interpretation:

- Choosing B requires an **additional investment of $5,000** at year 0,
- but yields **annual savings of $1,300**,
- with replacement and salvage effects included.

Step 3: Incremental IRR Equation (Using AW Form)

Let Δi be the **incremental rate of return**.

$$\begin{aligned} 0 = & -5{,}000(A/P, \Delta i, 12) \\ & +1{,}300 \\ & -11{,}000(P/F, \Delta i, 6)(A/P, \Delta i, 12) \\ & +3{,}000(A/F, \Delta i, 12) \end{aligned}$$

Solving this equation (trial-and-error or financial calculator):

$$\boxed{\Delta i^* = 11.2\%}$$

Step 4: Decision Rule (Incremental ROR)

- ΔIRR = **11.2%**
- MARR = **14%**

Since:

$$\Delta i^* < \text{MARR}$$

The **additional investment in B is not economically justified**.

Step 5: Final Decision

$$\boxed{\text{Select Alternative A}}$$

- Although Alternative B has **lower annual operating costs**,
- the **extra initial investment** does not earn a sufficient return.
- Incremental ROR focuses on **"Is the extra money worth it?"**

Benefit/Cost Analysis and Public Sector Economics

- Projects in private sector are owned by corporations, partnerships, and individuals and used by customers.
- Projects in public sector are owned, used and financed by citizens.
- Public sector projects have a primary purpose to provide services to the citizens for the public good at no profit.
  - Some public sector examples are:
  - Hospitals and clinics
  - Highways, bridges and waterways
  - Convention Centers
  - Schools, Colleges, Universities
  - Sports Arenas
- There are significant differences in the characteristics of private and public sector alternatives.

Characteristic	Public Sector	Private Sector
Size of Investment	Larger	Medium to Low
Life Estimates	Longer (30-50+ yrs)	Shorter (2-25 yrs)
Cash Flow Estimates	No Profit; cost, benefit, disbenefit	Revenues Costs
Funding	Taxes, bonds	Loans, stocks
Interest Rate	Lower	Higher
Alternative Selection Criteria	Multiple	ROR
Environment of Evaluation	Politically inclined	Economic

Cash Flow Estimates

- To perform an economic analysis of public alternatives, the costs (initial and annual), the benefits, and the disbenefits must be estimated as accurately as possible in monetary units.
- Costs: Estimated expenditures to the government entity for construction, operation, and maintenance of the project, less any expected salvage value.
- Benefits: Advantages to be experienced by the owners, the public
- Disbenefits: Expected undesirable or negative consequences to the owners, the public

- **Interest Rate for Public Sector Projects**
- The interest rate for the public sector is also identified as *i*; to distinguish it from the private sector rate called discount rate.

- All cost, benefit and disbenefit estimates must be converted to a common equivalent monetary unit (*PW*, *AW*, or *FW*) B/C ratio is then calculated as:

$$B/C = \frac{\text{PW of benefits}}{\text{PW of costs}} = \frac{\text{AW of benefits}}{\text{AW of costs}} = \frac{\text{FW of benefits}}{\text{FW of costs}}$$

- *PW* and *AW* are used more frequently
- Sign convention is + in B/C analysis
  - Costs are positive, salvage values are negative and subtracted from costs
  - Benefits are positive, disbenefits are negative and subtracted from benefits
- The decision guideline is:
  - If $B/C \geq 1.0$, accept the project
  - If $B/C < 1.0$, reject the project

- This ratio is called conventional B/C ratio (all terms in *PW* or *AW*):

$$B/C = \frac{Benefits - Disbenefits}{Cost - Salvage} = \frac{B - D}{C - S}$$

- The way how disbenefits handled changes the magnitude of the B/C ratio
  - Disbenefits are subtracted from benefits (correct)
  - Disbenefits are added to costs (incorrect)
- Handling the disbenefits incorrectly and adding them to costs does not alter the decision (B/C < or > 1)

$$\begin{aligned} B/C &= \frac{B - D}{C - S} < 1 \rightarrow B - D < C - S \\ &= B < C - S + D \rightarrow \frac{B}{C - S + D} < 1 \end{aligned}$$

- The modified B/C ratio includes maintenance and operation (M&O) costs in the numerator and treats them similar to disbenefits
- Once all terms are expressed in *PW*, *AW*, or *FW* terms:

$$\textit{Modified } B/C = \frac{\textbf{Benefits - Disbenefits-M\&O costs}}{\textbf{Initial Investment-Salvage}}$$

- The modified B/C ratio yields a different cost then conventional B/C ratio
- But the decision to accept or reject is same with both ratios
- There is also the benefit and cost difference measure of worth
- Based on the difference of *PW* or *AW* of benefits and costs
  - (B-C)=Net Benefits - (Costs-Salvage)
- The decision guideline is:
  - If $(B - C) \geq 0$, accept the project
  - If $(B - C) < 0$, reject the project

EXAMPLE

- The Ford Foundation expects to award $15 million today in grants to public high schools to develop new ways to teach the fundamentals of engineering that prepare students for university-level material. The grants will extend over a 10-year period and will create an estimated savings of $1.5 million per year in faculty salaries. The foundation uses a 6% discount rate for all grant awards. This grant program will share the Foundation funding with ongoing activities, so an estimated $200,000 per year will be removed from other program funding. Also $500,000 per year will be incurred as annual operation costs. Use the B/C method to determine if the grants program is economically justified using:

  1 Conventional B/C analysis

  2 Modified B/C ratio

  3 benefit and cost difference measure of wort

Use *AW* as the common monetary equivalent:

- *AW* of investment cost $= 15,000,000(A/P, 6\%, 10) = \$2,038,050$
- *AW* of benefit = \$1,500,000
- *AW* of disbenefit = \$200,000
- *AW* of M&O cost = \$500,000

1. Conventional B/C

$$B/C = \frac{\$1,500,000 - \$200,000}{\$2,038,050 + \$500,000} = 0.51 < 1$$

2. Modified B/C

$$Modified\ B/C = \frac{\$1,500,000 - \$200,000 - \$500,000}{\$2,038,050} = 0.39 < 1$$

3. $(B - C) = (1,500,000 - 200,000) - (2,038,050 + 500,000) \leq 0$

Program is not justified.

- The technique to compare two mutually exclusive alternatives using B/C ratio is the incremental B/C analysis. The procedure for incremental B/C is given as (equivalent values can be in *PW* or *AW*):

  1 Determine the total equivalent costs for both alternatives

  2 Order the alternatives by the total equivalent cost, smaller first. Calculate the incremental cost (*C*) for the larger-cost alternative over the smaller-cost alternative.

  3 Calculate the total equivalent benefits and disbenefits for both alternatives. Calculate incremental net benefits ((*B* − *D*)) for the larger-cost alternative over the smaller-cost alternative.

  4 Calculate the incremental B/C ratio

$$\Delta B/C = \frac{\Delta(B-D)}{\Delta C}$$

- Select the higher-cost alternative if $\Delta B/C \geq 1$

EXAMPLE

In the past, the Afram Foundation has awarded many grants to improve the living and medical conditions of people in war-torn and poverty-stricken countries throughout the world. In a proposal for the foundation's board of directors to construct a new hospital and medical clinic complex in a deprived central African country, the project manager has developed some estimates. These are developed, so she states, in a manner that does not have a major negative effect on prime agricultural land or living areas for citizens.

- Award amount: $20 million (end of) first year, decreasing by $5 million per year for 3 additional years; local government will fund during the first year only
- Annual costs: $2 million per year for 10 years, as proposed
- Benefits: Reduction of $8 million per year in health-related expenses for citizens
- Disbenefits: $0.1 to $0.6 million per year for removal of arable land and commercial districts

Use the conventional and modified B/C methods to determine if this grant proposal is economically justified over a 10-year study period. The foundation's discount rate is 6% per year.

- Initially, determine the AW for each parameter over 10 years. In $1 million units,
- Award: 20 − 5(*A*/*G*,6%,4) = $12.864 per year
- Annual costs: $2 per year
- Benefits: $8 per year
- Disbenefits: Use $0.6 for the first analysis

The *conventional B/C analysis* applies

$$\text{B/C} = \frac{8.0 - 0.6}{12.864 + 2.0} = 0.50$$

The *modified B/C analysis* uses

$$\text{Modified B/C} = \frac{8.0 - 0.6 - 2.0}{12.864} = 0.42$$

- The proposal is not justified economically since both measures are less than 1.0. If the low disbenefits estimate of $0.1 million per year is used, the measures increase slightly, but not enough to justify the proposal.

EXAMPLE

- A municipality is considering building a **new public bus lane** to reduce traffic congestion and pollution. The project has the following estimated impacts:

Costs

- Initial construction cost: **$18 million** (year 0)
- Annual maintenance cost: **$1.2 million per year** for 20 years

Benefits

- Reduced travel time for commuters: **$2.8 million per year**
- Reduced vehicle operating costs: **$0.9 million per year**
- Environmental benefits (lower emissions, noise): **$0.5 million per year**

Other Information

- Project life: **20 years**
- Discount rate (social discount rate): **6% per year**
- No salvage value

$$\text{Annual Benefits} = 2.8 + 0.9 + 0.5 = 4.2 \text{ million dollars/year}$$

$$PW_B = 4.2 \times (P/A, 6\%, 20)$$

$$(P/A, 6\%, 20) = 11.47$$

$$PW_B = 4.2 \times 11.47 = 48.17 \text{ million}$$

Initial Cost

$$PW_{C,0} = 18$$

Maintenance Costs

$$PW_{C,maint} = 1.2 \times (P/A, 6\%, 20)$$

$$PW_{C,maint} = 1.2 \times 11.47 = 13.76$$

Total Present Worth of Costs

$$PW_C = 18 + 13.76 = 31.76 \text{ million}$$

$$B/C = \frac{PW_B}{PW_C} = \frac{48.17}{31.76} = 1.52$$

- If **B/C > 1**, the project is economically justified.
- If **B/C < 1**, the project should be rejected.

Since: $\boxed{B/C = 1.52 > 1}$

The bus lane project should be accepted.

Cautions in using Incremental B/C Analysis

1 Each alternative included in the incremental B/C analysis must have a B/C ratio ≥ 1.

2 Ordering of alternatives is done on the basis of total cost not initial Cost

3 Like all methods considered so far, incremental B/C analysis requires equal-service comparison of alternatives.

- Usually the alternatives have very long lives, so alternatives have equal lives
- When alternatives do have unequal lives, LCM of lives must be used
- Use *AW* equivalency of costs and benefits whenever alternatives have unequal lives

EXAMPLE

The city of Garden Ridge, Florida, has received designs for a new patient room wing to the municipal hospital from two architectural consultants. The three financial estimates are given as:

	Design A	Design B
Construction Cost, $	$1,000,000	$1,500,000
Building maintenance cost, $ per year	35,000	55,000
Savings in operation, $ per year	200,000	450,000

The discount rate is 5%, and the life of the building is estimated as 30 years. Use conventional B/C ratio analysis to select design A or B.

Use *AW* equivalency:

1. *AW* of costs is the sum of construction and building maintenance costs

$$AW_A = 1,000,000(A/P, 5\%, 30) + 35,000 = \$100,050$$
$$AW_B = 1,500,000(A/P, 5\%, 30) + 55,000 = \$152,575$$

2. Design B is the alternative with the larger total cost, so it is the alternative to be incrementally justified.

$$\Delta C = 152,575 - 100,050 = \$52,525$$

3. Savings in operation are the benefits:

$$\Delta B = 450,000 - 200,000 = \$250,000$$

4. The incremental B/C ratio is calculated as:

$$\Delta B/C = \frac{\$250,000}{\$52,525}$$

5. The incremental B/C is larger than 1, Therefore design B is selected.

for Multiple Mutually Exclusive Alternatives

1. Determine the the total equivalent costs for all alternatives (Use *PW* or *AW*)
2. Order the alternatives by the total equivalent cost, smaller first
3. Calculate the total equivalent benefits and disbenefits for each alternative
4. Direct benefit estimation only: Determine the B/C for all alternatives and retain the ones with $B/C \geq 1$
5. Calculate the incremental costs (ΔC) and benefits ($\Delta(B - D)$):
   - $\Delta C =$ challenger cost - defender cost
   - $\Delta(B - D) =$ challenger benefits - defender benefits
   - if relative usage costs are estimated for each alternative, rather than direct benefits:
     - $\Delta(B - D) =$ defender usage cost - challenger usage cost

6. Calculate the incremental B/C for the first challenger compared to the defender
   - if $\Delta B/C \geq 1$, eliminate the defender, challenger becomes new defender
   - if $\Delta B/C < 1$, eliminate the challenger
7. Repeat steps 5 and 6 until only one alternative remains

Example

A municipal is seeking a developer that will place a major water park in the city area. Financial incentives will be awarded to the developers depending on their investment size. The developers will get reductions in property tax for 8 years. Each proposal includes a provision that residents of the county will benefit from reduced entrance fees when using the park. This reduction will be effective for 8 years. The annual total entrance fees with the reduction are estimated. The discount rate is 7 %. Which proposal should be selected?

	Proposal 1	Proposal 2	Proposal 3
Initial Incentive, $	$250,000	$350,000	500,000
Tax Incentive cost, $ per year	25,000	35,000	50,000
Entrance fees, $ per year	500,000	450,000	425,000
Extra sales tax, $ per year	310,00	320,000	320,000
Study period, years	8	8	8

- *AW* of costs is the sum of incentive cost and the annual cost of initial incentive.

$$\begin{aligned} AW_{C1} &= 250,000(A/P,7\%,8) + 25,000 = \$66,867 \\ AW_{C2} &= 350,000(A/P,7\%,8) + 35,000 = \$93,614 \\ AW_{C3} &= 500,000(A/P,7\%,8) + 50,000 = \$133,735 \end{aligned}$$

  Alternatives are ordered as 1,2,3.

- The annual benefits are the incremental benefit from the decrease in entrance fees and increase in sales taxes.

$$\begin{aligned} \Delta B_{2-1} &= 50,000 + 10,000 = \$60,000 \\ \Delta B_{3-2} &= 25,000 + 0 = \$25,000 \end{aligned}$$

The first defender is alternative 1.

- For 2-1 comparison:

$$\Delta B/C = \frac{\Delta B}{\Delta C} = \frac{\$60,000}{93,614 - 66,867} > 1$$

Alternative 2 is incrementally justified. Alternative 1 is eliminated, 2 is the new defender.

- For 3-2 comparison:

$$\Delta B/C = \frac{\Delta B}{\Delta C} = \frac{\$25,000}{133,735 - 93,614} < 1$$

Alternative 3 is incrementally not justified. Alternative 3 is eliminated, Alternative 2 is the selection.

EXAMPLE

- The city of Garden Ridge, Florida, has received designs for a new patient room wing to the municipal hospital from two architectural consultants. One of the two designs must be accepted in order to announce it for construction bids. The costs and benefits are the same in most categories, but the city financial manager decided that the estimates below should be considered to determine which design to recommend at the city council meeting next week and to present to the citizenry in preparation for an upcoming bond referendum next month.

	Design A	Design B
Construction cost, $	10,000,000	15,000,000
Building maintenance cost, $/year	35,000	55,000
Patient usage copay, $/year	450,000	200,000

- The patient usage copay is an estimate of the amount paid by patients over the insurance coverage generally allowed for a hospital room. The discount rate is 5%, and the life of the building is estimated at 30 years.

(*a*) Use incremental B/C analysis to select design A or B.

(*b*) Once the two designs were publicized, the privately owned hospital in the directly adjacent city of Forest Glen lodged a complaint that design A will reduce its own municipal hospital' income by an estimated $500,000 per year because some of the day-surgery features of design A duplicate its services. Subsequently, the Garden Ridge merchants' association argued that design B could reduce its annual revenue by an estimated $400,000 because it will eliminate an entire parking lot used by their patrons for short-term parking. The city financial manager stated that these concerns would be entered into the evaluation as disbenefits of the respective designs. Redo the B/C analysis to determine if the economic decision is still the same as when disbenefits were not considered.

(*a*) Since most of the cash flows are already annualized, the incremental B/C ratio will use AW values. No disbenefit estimates are considered. Follow the steps of the procedure above:

1. The AW of costs is the sum of construction and maintenance costs.

$$AW_A = 10{,}000{,}000(A/P,5\%,30) + 35{,}000 = \$685{,}500$$
$$AW_B = 15{,}000{,}000(A/P,5\%,30) + 55{,}000 = \$1{,}030{,}750$$

2. Design B has the larger AW of costs, so it is the alternative to be incrementally justified. The incremental cost is

$$\Delta C = AW_B - AW_A = \$345{,}250 \text{ per year}$$

3. The AW of benefits is derived from the patient usage copays, since these are consequences to the public. The benefits for the ΔB/C analysis are not the estimates themselves, but the *difference* if design B is selected. The lower usage copay is a positive benefit for design B.

$$\Delta B = \text{copay}_A - \text{copay}_B = \$450{,}000 - \$200{,}000 = \$250{,}000 \text{ per year}$$

4. The incremental B/C ratio is calculated by Equation [9.2].

$$\Delta B/C = \frac{\$250{,}000}{\$345{,}250} = 0.72$$

5. The B/C ratio is less than 1.0, indicating that the extra costs associated with design B are not justified. Therefore, design A is selected for the construction bid.

(*b*) The revenue loss estimates are considered disbenefits. Since the disbenefits of design B are $100,000 less than those of A, this positive difference is added to the $250,000 benefits of B to give it a total benefit of $350,000. Now

$$\Delta B/C = \frac{\$350{,}000}{\$345{,}250} = 1.01$$

Design B is slightly favored. In this case, the inclusion of disbenefits has reversed the previous economic decision. This has probably made the situation more difficult politically. New disbenefits will surely be claimed in the near future by other special-interest groups.